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Project Title	Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions
Week	Week 6

WEEK-6 Technical Report

Task Name

Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions

Abstract

This project focuses on analyzing **15 million user path logs** using a **Markov Chain Attribution Model**. A transition probability matrix is created to represent customer interactions across different marketing channels. The **Removal Effect** method is used to measure the contribution of each channel by calculating the reduction in conversion probability when a channel is removed. The results help identify the most effective marketing channels and improve marketing strategies.

1.Introduction

In digital marketing, customers often interact with multiple channels before making a purchase or conversion. Understanding the contribution of each channel is important for effective marketing decisions. This project uses a **High-Dimensional Markov Chain Model** to analyze customer journeys and calculate channel importance. By applying the **Removal Effect** technique, the impact of each marketing channel on conversion rates can be accurately measured and ranked.

2. Objectives

1. To analyze large-scale user journey data containing 15 million path logs.
2. To construct a transition probability matrix for marketing channels.
3. To model customer journeys using Markov Chains.
4. To calculate the baseline conversion probability.

- 5. To compute the Removal Effect for each marketing channel.
- 6. To identify the most influential channels in the conversion process.
- 7. To provide data-driven insights for marketing budget allocation.
- 8. To improve marketing campaign performance through accurate attribution.

3. Problem Statement

Organizations invest heavily in multiple marketing channels such as Social Media, Email Marketing, Search Ads, Display Ads, and Direct Traffic. Traditional attribution methods often provide inaccurate channel credit. Therefore, an advanced attribution model is required to determine the actual contribution of each channel toward customer conversions.

4. System Architecture

User Path Logs (15 Million Records)



Data Preprocessing



Transition Count Matrix



Transition Probability Matrix



Markov Chain Model



Baseline Conversion Probability





Channel Removal Analysis



Removal Effect Index



Channel Attribution Ranking

5. Methodology

Step 1: Data Collection

Collect 15 million user path logs containing customer interactions with various marketing channels.

Step 2: Data Preprocessing

Clean the dataset by removing duplicate records, handling missing values, and standardizing channel names.

Step 3: Transition Matrix Construction

Count transitions between marketing channels and create a transition probability matrix.

Step 4: Markov Chain Modeling

Represent customer journeys as a Markov Chain where each marketing channel is considered a state.

Step 5: Baseline Conversion Calculation

Calculate the probability of users reaching the conversion state using the complete transition matrix.

Step 6: Removal Effect Analysis

Remove one channel at a time and recalculate the conversion probability.

Step 7: Attribution Score Generation

Calculate the percentage reduction in conversion probability and assign attribution scores.

6. Removal Effect Formula

The Removal Effect is calculated using:

$$\text{Removal Effect} = \frac{\text{Pbaseline} - \text{Premoved}}{\text{Pbaseline}} \times 100$$

Where:

- P_{baseline} = Original conversion probability
- P_{removed} = Conversion probability after removing a channel

7. Expected Results

- Transition probability matrix for all channels.
- Baseline conversion probability.
- Removal Effect values for each channel.
- Ranking of marketing channels by contribution.
- Visualization of channel importance.

8. Applications

- Digital Marketing Analytics
- Customer Journey Analysis
- Marketing Budget Optimization
- Campaign Performance Measurement
- Customer Behavior Modeling
- Business Intelligence Systems.

9. Advantages

- Provides accurate channel attribution.
- Considers the entire customer journey.
- Supports large-scale data analysis.
- Improves marketing ROI.
- Enables data-driven decision making.

10. Conclusion

The **Multi-Channel Marketing Attribution with High-Dimensional Markov Chain Transitions** project provides an advanced approach for evaluating marketing channel effectiveness. By constructing a transition probability matrix from 15 million user path logs and calculating the Removal Effect for each channel, organizations can accurately determine channel contributions and optimize marketing investments. This approach offers a more reliable alternative to traditional attribution models and enhances overall marketing performance.

